Step-by-step guide

Monthly Production Planning Using the Refinery Planning Workbook (Manual Mode)

1. Purpose

This guide describes how a production planner updates, builds and checks a monthly refinery production plan in the planning workbook **without** the Python optimizer. Following it gives a plan that respects every unit, quality and tank constraint, and a margin figure that can be compared against the optimised benchmark.

2. Scope

Covers a 31-day horizon with four crudes (Arab Light, Murban, Tapis, Kimanis), the CDU, Reformer, RFCC and DHT, and ten tank groups (4 crude, 6 product). Covers mid-month re-planning (rolling forward actual days). Does not cover running the Python optimizer, crude selection/purchase decisions or unit shutdown planning.

3. Roles & Responsibilities

Role	Responsibility in this guide
Production Planner	Owns the workbook. Updates inputs, builds the plan, clears all violations, issues the plan.
Senior Planner / Planning Lead	Reviews the plan using the Section 9 checklist.
Production Accounting / Oil Movement	Supplies agreed opening tank stocks and daily actuals.
Crude Supply & Marketing	Supplies crude arrival schedule, product lifting schedule and price deck.
Process Engineering / Operations	Confirms unit capacities, turndowns, yields and any constraint changes for the month.
Laboratory / Crude Assay	Supplies crude yields and residue sulfur.

4. Definitions

Term	Meaning
kbbl, kbbl/d	Thousand barrels; thousand barrels per day. All volumes in the workbook use these units.
Actual day	A day already run. Values are fixed history from production accounting (grey rows).
Plan day	A future day whose rates the planner decides (green cells).
Decision cells	The six cells per plan day the planner sets: four crude rates, Reformer feed, RFCC feed (Daily_Plan D:I).
Min heel	Lowest allowed tank inventory (pump suction / operating minimum).
VIOL / FEASIBLE	Constraint_Check result: a breached constraint, or a day/plan with no breaches.
RON	Research Octane Number of the gasoline pool. Spec is 95 minimum.

5. Workbook Map

Work through the sheets in the order below. Inputs first, then actuals, then the plan, then the checks.

Order	Sheet	Type	What it holds
1	Crude_Data	Input	Crude prices, cut yields, residue sulfur, crude rate limits
2	Unit_Data	Input	Unit capacities, turndowns, ramp limit, opex, unit yields, RON & sulfur specs
3	Product_Data	Input	Product netback prices
4	Tank_Master	Input	Tank capacity, min heel, opening stock (1st of month), end-of-month target
5	Crude_Receipts	Input	Daily crude cargo arrivals by grade
6	Product_Liftings	Input	Daily product liftings and vessel cargoes
7	Daily_Plan	Plan	Actual rows + plan rows; all yields, product make, quality and margin; KPIs
8	Tank_Summary	Check	Closing stock and fill % per tank per day, with charts
9	Tank_Balance	Check	One row per tank per day: opening + in − out = closing, with status
10	Constraint_Check	Check	OK/VIOL per constraint per day; total violations in X37
–	Optimization_Notes	Reference	Objective function, constraints C1–C16, assumptions, optimum benchmark

Cell colour legend

Blue text, yellow fill	Input – edit this
Blue text, green fill	Plan decision cell – planner sets this
Blue text, grey fill	Actual day – enter actuals only, do not plan
Black text	Formula – never overwrite
Green text	Link to another sheet – never overwrite

Golden rule: only type into blue-text cells. Do not insert or delete rows or columns in any sheet – formulas reference fixed row numbers (especially Unit_Data and Tank_Master). Save a dated copy before each planning cycle.

6. Procedure

Step 1 – Update crude data (Crude_Data)

#	Action	Where	Check / acceptance
1.1	Enter the month's delivered crude prices.	Row 4, C:F	Price deck date matches the plan month.
1.2	Enter cut yields (LPG, naphtha, kerosene, gasoil, residue, loss) from the latest assay.	Rows 5–10	–
1.3	Check the yield closure row.	Row 11	Each crude = **1.000**. If not, fix yields before going on.
1.4	Enter residue sulfur per crude.	Row 12	Matches assay / recent lab results.
1.5	Confirm max and min CDU rate per crude agreed with Operations.	Rows 13–14	Any temporary limits (e.g. corrosion, wax) are reflected.

Why 1.3 matters: a yield set that does not close to 1.000 creates or loses barrels silently. Margins and tank balances will look plausible but be wrong.

Step 2 – Update unit data (Unit_Data)

#	Action	Where	Check / acceptance
2.1	Confirm CDU max/min throughput and day-to-day ramp limit.	D4:D6	Agreed with Operations for the month.
2.2	Confirm Reformer, RFCC and DHT max/min feed.	D8:D9, D14:D15, D22:D23	Reduce max if a unit is constrained (catalyst, fouling, partial outage).
2.3	Update variable opex.	D7, D10, D16, D24	From latest cost data.
2.4	Confirm unit yields and blend RON values.	D11:D12, D17:D20, D25, D26:D27	Calculated balance rows (D13, D21) are ≥ 0.
2.5	Confirm product specs: gasoline RON min, residue sulfur max.	D28:D29	Matches current product specification.

Step 3 – Update product prices (Product_Data)

#	Action	Where	Check / acceptance
3.1	Enter product netback prices from Marketing.	Row 4, C:H	Same price-deck date as crude prices.

Step 4 – Set opening position and logistics

#	Action	Where	Check / acceptance
4.1	Enter agreed opening stock for 1st of month per tank.	Tank_Master F4:F13	Matches signed-off month-end stock report.
4.2	Confirm working capacity and min heel per tank.	Tank_Master D:E	Reflect any tank out of service.
4.3	Enter end-of-month inventory targets.	Tank_Master G4:G13	Agreed with Supply/Marketing for next-month start.
4.4	Enter crude cargo arrivals by date and grade.	Crude_Receipts C5:F35	Total row 36 matches the shipping programme.
4.5	Enter daily product liftings and vessel cargoes.	Product_Liftings C5:H35	Total row 36 matches the Marketing commitment.

Receipts and liftings are **fixed** in this workbook. The plan must fit around them. If a schedule is impossible to meet, raise it with Supply/Marketing – do not change the schedule just to clear a violation.

Step 5 – Enter actual days (Daily_Plan)

#	Action	Where	Check / acceptance
5.1	For each day already run, enter actual crude rates, Reformer feed and RFCC feed.	Daily_Plan D:I (grey rows)	Values from production accounting, not plan.
5.2	Make sure the Status column reads 'Actual' for these days and 'Plan' for the rest.	Daily_Plan column C	Cut-off day = last fully accounted day.
5.3	Compare calculated stock at the cut-off day against the physical tank dip.	Tank_Summary	Difference within normal accounting tolerance. Investigate if not.

Step 6 – Build the plan for the remaining days (Daily_Plan)

Only the six green cells per plan day are set (columns D–I). Set them in the order below – each one limits the next.

Step 6a – Check crude availability first

#	Action	Where	Check / acceptance
6.1	Look at the crude tank lines in Tank_Summary up to each cargo arrival.	Tank_Summary C:F, N:Q	Note which crude runs short before its next cargo.
6.2	Pencil in the maximum each crude can run until its next arrival without going below min heel.	–	This sets the upper limit on that crude, before margin.

Step 6b – Set total CDU rate

#	Action	Where	Check / acceptance
6.3	Choose the daily CDU total.	Daily_Plan J (result)	Between CDU min and max (100–150 kbbl/d).
6.4	Check day-to-day change, including actual last day → first plan day.	Constraint_Check H:I	Change ≤ ramp limit (±10 kbbl/d).

Step 6c – Split the crude slate

#	Action	Where	Check / acceptance
6.5	Distribute the CDU total across the four crudes.	Daily_Plan D:G	Each crude within its own max (Crude_Data row 13).
6.6	Check residue sulfur.	Daily_Plan P	≤ 1.50 wt%.

Rule of thumb (sulfur): each 1 kbbl/d of Arab Light needs about 1.7–1.9 kbbl/d of Tapis or Kimanis to keep the residue pool at 1.5 wt%. Murban helps very little. Arab Light usually earns the most margin, so sulfur – not price – limits how much you run.

Step 6d – Set Reformer and RFCC feed

#	Action	Where	Check / acceptance
6.7	Set RFCC feed. Default to the maximum if residue is available.	Daily_Plan I	≤ atmospheric residue made (col O). Within RFCC limits.
6.8	Set Reformer feed. Default to the maximum if naphtha is available.	Daily_Plan H	≤ SR naphtha made (col L). Within Reformer limits.
6.9	Check gasoline RON.	Daily_Plan AD	≥ 95.0.

Rule of thumb (octane): Reformer feed ≥ about 0.62 × RFCC feed keeps the pool at 95 RON. At RFCC 35 kbbl/d, Reformer feed must be at least about 21.7 kbbl/d.

Step 6e – Check DHT loading

#	Action	Where	Check / acceptance
6.10	Review DHT feed (SR gasoil + LCO). It is calculated, not set directly.	Daily_Plan W	Between 20 and 48 kbbl/d. Too high → reduce gasoil-rich crudes or RFCC feed.

Step 7 – Verify the plan (Constraint_Check, Tank_Balance)

#	Action	Where	Check / acceptance
7.1	Review violations per constraint.	Constraint_Check row 37	All counts = 0.
7.2	Clear unit and quality breaches first (C1–C10).	Constraint_Check E:U	Usually fixed by a rate or slate change.
7.3	Clear tank breaches next (C13). Filter Status for BELOW MIN / ABOVE MAX.	Tank_Balance column L	No breaches on any day.
7.4	Clear end-of-month targets last (C14).	Constraint_Check A42:E51, E52	All OK. E52 = 0.
7.5	Confirm overall status.	Constraint_Check X37; Daily_Plan G43:G44	X37 = 0 and status shows **FEASIBLE**.

Fixing common violations

Symptom	Likely cause	Typical fix
C1 / C3 VIOL – CDU rate or ramp	Total too high/low, or a big step between days	Spread the change over several days.
C5 / C7 VIOL – feed > availability	Reformer/RFCC feed set above naphtha/residue made	Lower feed, or move slate to crudes richer in that cut.
C9 VIOL – RON below 95	Too much FCC gasoline vs reformate	Raise Reformer feed or lower RFCC feed.
C10 VIOL – residue sulfur	Too much Arab Light	Swap Arab Light for Tapis/Kimanis.
Crude tank BELOW MIN	Running a grade too hard before its cargo	Cut that grade earlier; swap to another crude.
Product tank ABOVE MAX	Production ahead of a lifting	Lower the unit that fills it for a few days, or raise a lifting with Marketing.
Product tank BELOW MIN	Big lifting before stock has built	Build stock earlier; check with Marketing on lifting date.
C14 VIOL – month-end target	Stock drawn down late in month	Adjust the last 5–10 days; check targets are realistic.

Step 8 – Review economics

#	Action	Where	Check / acceptance
8.1	Read plan-days gross margin (the planner's objective).	Daily_Plan G41	Compare with previous plan revision.
8.2	Read month margin and margin per bbl crude.	Daily_Plan G39, AI36	Reasonable vs last month; explain big changes.
8.3	Compare against the optimised benchmark.	Optimization_Notes, Section 9	Gap noted in plan memo. A well-built manual plan is usually within a few percent.
8.4	Sense-check unit loadings.	Daily_Plan H, I	Reformer and RFCC normally at or near max unless a spec/tank limits them – record the reason if not.

7. Mid-Month Re-Planning (Rolling Forward)

- Save a new dated copy of the workbook.
- For each newly completed day, change Status (column C) from 'Plan' to 'Actual' and overwrite D:I with accounted actuals.
- Update Crude_Receipts and Product_Liftings for any schedule changes.

- Re-check Tank_Summary at the new cut-off day against physical stock (Step 5.3).
- Re-run Steps 6–8 for the remaining plan days only.

Note: cell fill colours are cosmetic. The Status column is what separates actual days from plan days in the KPIs.

8. New Month Set-Up

- Copy last month's workbook and rename it for the new month.
- Update the date in Daily_Plan B5 and the following date cells to the new month (other sheets link to these).
- Set opening stock in Tank_Master F to the signed-off month-end stock of the previous month.
- Reset all Status cells to 'Plan', then follow Steps 1–8.
- The workbook is built for 31 days and checks month-end targets on day 31. For shorter months, treat the extra rows as the first days of next month and keep them feasible; read month-end stock from the true last day in Tank_Summary.

9. Plan Review Checklist

✓	Check	Notes
☐	Price deck and assay dates confirmed (Crude_Data, Product_Data)	
☐	Yield closure = 1.000 for every crude (Crude_Data row 11)	
☐	Unit limits agreed with Operations (Unit_Data)	
☐	Opening stock matches signed-off stock report (Tank_Master F)	
☐	Receipts and liftings match latest schedules	
☐	Actual days entered and Status column correct	
☐	Constraint_Check X37 = 0 and Daily_Plan status = FEASIBLE	
☐	End-of-month targets met (Constraint_Check E52 = 0)	
☐	Margin reviewed and gap to benchmark recorded	
☐	Dated copy saved and plan shared	

10. References

- Optimization_Notes sheet – objective function, constraints C1–C16, assumptions and benchmark.
- Note: all prices, yields and capacities in the delivered workbook are illustrative test values and must be replaced with site data before use.